ClipSAT : Differential Equations First/Second Order, Laplace, Systems & S

Unit 6.01: The Laplace Transform, Step Forcing & Convolution

Name: _____

Q1. Find mathcal L{3 t + cos{leftt right}} .

[ ] A: frac{2 s^{3} + 3 s^{2} + s + 3}{s^{4

[ ] B: frac{- s^{3} - 3 s^{2} - 3}{s^{4} +

[ ] C: frac{s^{3} + 3 s^{2} + 3}{s^{3} + s}

[ ] D: frac{s^{3} + 3 s^{2} + 3}{s^{4} + s^

Q2. Using the definition, mathcal L{e^{2 t}}=int_0^infty e^{-s-2t},dt equals for s>2 :

[ ] A: frac{s}{s - 2}

[ ] B: frac{1}{s - 2}

[ ] C: frac{1}{s + 2}

[ ] D: - frac{1}{s - 2}

Q3. If mathcal L{y}=Ys , y0=-3 and y'0=-1 , find mathcal L{y''} .

[ ] A: Y s^{2} + s + 3

[ ] B: Y s^{2} - 3 s - 1

[ ] C: Y s^{2} + 3 s + 1

[ ] D: Y s + 4

Q4. Find mathcal L{t^{4}} .

[ ] A: frac{24}{s^{4}}

[ ] B: frac{24}{s^{5}}

[ ] C: frac{4}{s^{5}}

[ ] D: frac{24}{s^{6}}

Q5. Using the definition, mathcal L{2 t}=int_0^infty 2te^{-st},dt equals for s>0 :

[ ] A: frac{2}{s}

[ ] B: frac{2}{s^{3}}

[ ] C: frac{2}{s^{2}}

[ ] D: - frac{2}{s^{2}}

Q6. The second-order linear homogeneous ODE f''x - 53f'x + 696fx = 0 has characteristic roots lambda_1, lambda_2 . What is the trace sum lambda_1 + lambda_2 ?

[ ] A: 53

[ ] B: 56

[ ] C: 77

[ ] D: 28

Q7. If mathcal L{y}=Ys and y0=1 , find mathcal L{y'} .

[ ] A: Y s - 1

[ ] B: Y s + 1

[ ] C: Y - 1

[ ] D: Y s

Q8. Find mathcal L{t^{3}} .

[ ] A: frac{6}{s^{4}}

[ ] B: frac{6}{s^{3}}

[ ] C: frac{3}{s^{4}}

[ ] D: frac{6}{s^{5}}